

P1.

$$[Q(s, a) - V(s)] =$$

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①

$$\begin{aligned} L_{\pi_{\theta_1}}(\pi_{\theta}) &= \eta(\pi_{\theta_1}) + \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \sum_a \pi_{\theta}(a|s) A^{\pi_{\theta_1}}(s, a) \\ &= \eta(\pi_{\theta_1}) + \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \left\{ \sum_a [\pi_{\theta}(a|s) Q(s, a)] - \sum_a [\pi_{\theta}(a|s) Q(s, a)] \right\} \\ &= \eta(\pi_{\theta_1}) + 0 = \eta(\pi_{\theta_1}). \end{aligned}$$

by  $V(s) = \sum_a \pi_{\theta}(a|s) Q(s, a)$ ,  $\sum_a \pi_{\theta}(a|s) = 1$

②

by likelihood ratio trick:

$$\nabla_{\theta} \eta(\pi_{\theta}) = \sum_s d_{\mu}^{\pi_{\theta}}(s) \sum_a \pi_{\theta}(a|s) \nabla_{\theta} \log \pi_{\theta}(a|s) \cdot Q^{\pi_{\theta}}(s, a).$$

$$\nabla_{\theta} (\eta(\pi_{\theta}) - \eta(\pi_{\theta_1})) = \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \sum_a \pi_{\theta}(a|s) \nabla_{\theta} \log \pi_{\theta}(a|s) \cdot \hat{A}^{\pi_{\theta_1}}(s, a).$$

$$\nabla_{\theta} \eta(\pi_{\theta})|_{\theta=\theta_1} = \nabla_{\theta} [\eta(\pi_{\theta}) - \eta(\pi_{\theta_1})]|_{\theta=\theta_1} = \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \sum_a \pi_{\theta_1}(a|s) \nabla_{\theta} \log \pi_{\theta}(a|s) \cdot \hat{A}^{\pi_{\theta_1}}(s, a).$$

by the same way

$$\nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta}) = \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \sum_a \pi_{\theta}(a|s) \nabla_{\theta} \log \pi_{\theta}(a|s) \cdot \hat{A}^{\pi_{\theta_1}}(s, a)$$

$$\nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta})|_{\theta=\theta_1} = \sum_s d_{\mu}^{\pi_{\theta_1}}(s) \sum_a \pi_{\theta_1}(a|s) \nabla_{\theta} \log \pi_{\theta_1}(a|s) \cdot \hat{A}^{\pi_{\theta_1}}(s, a).$$

P2.

① to solve  $D(\lambda)$ , consider gradient of  $\theta$ .

\* 和  $\theta$  無關!

$$\nabla_{\theta} L(\theta, \lambda) = -(\nabla_{\theta} L_{\theta_k}(\theta))|_{\theta=\theta_k} + \lambda H_{\theta_k} \theta = 0$$

$$\Rightarrow \Delta \theta = \frac{1}{\lambda} H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}), \text{ denote } g = \nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}, \text{ then } \Delta \theta = \frac{1}{\lambda} H_{\theta_k}^{-1} g.$$

$$\begin{aligned} D(\lambda) &= -g^T (\frac{1}{\lambda} H_{\theta_k}^{-1} g) + \frac{1}{2} (\frac{1}{\lambda} H_{\theta_k}^{-1} g)^T H (\frac{1}{\lambda} H_{\theta_k}^{-1} g) = -\frac{1}{\lambda} g^T H^{-1} g + \frac{1}{2\lambda} g^T H^{-1} g - \lambda \delta \\ &= -\frac{1}{2\lambda} g^T H^{-1} g - \lambda \delta = -\frac{1}{2\lambda} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k})^T H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}) - \lambda \delta. \end{aligned}$$

②

$$\frac{d}{d\lambda} D = -\frac{1}{2\lambda^2} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k})^T H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}) - \delta, \quad (L=0)$$

$$\Rightarrow \lambda^* = \left( \frac{(\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k})^T H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k})}{2\delta} \right)^{\frac{1}{2}}$$

$$\text{we know } \Delta \theta = \theta^* - \theta_k = \frac{1}{\lambda} H^{-1} g, \quad \theta^* = \theta_k + \frac{1}{\lambda} H^{-1} g.$$

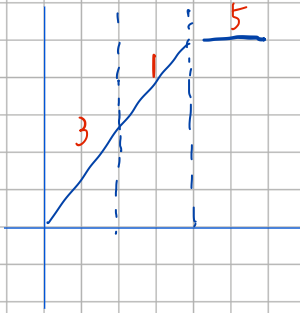
$$\theta^* = \theta_k + \Delta H^{-1} g, \quad \lambda = \frac{1}{\lambda} = \left( \frac{\delta}{(\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k})^T H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k})} \right)^{\frac{1}{2}}$$

P3.

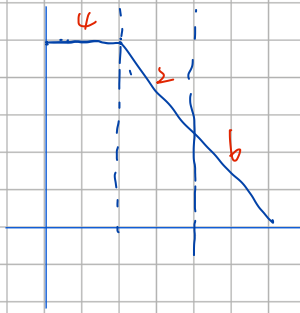
①  $L_{s,a}^{\text{clip}}(\theta; \theta_k)$

| $P_t(\theta) > 0$                            | $A_t$ | Return of min      | objective clipped | sign of objective | Gradient. |
|----------------------------------------------|-------|--------------------|-------------------|-------------------|-----------|
| 1 $P_t(\theta) \in [1-\epsilon, 1+\epsilon]$ | +     | $P_t(\theta) A_t$  | x                 | +                 | ✓         |
| 2 $P_t(\theta) \in [1-\epsilon, 1+\epsilon]$ | -     | $P_t(\theta) A_t$  | x                 | -                 | ✓         |
| 3 $P_t(\theta) < 1-\epsilon$                 | +     | $P_t(\theta) A_t$  | x                 | +                 | ✓         |
| 4 $P_t(\theta) < 1-\epsilon$                 | -     | $(1-\epsilon) A_t$ | ✓                 | -                 | 0         |
| 5 $P_t(\theta) > 1+\epsilon$                 | +     | $(1+\epsilon) A_t$ | ✓                 | +                 | 0         |
| 6 $P_t(\theta) > 1+\epsilon$                 | -     | $P_t(\theta) A_t$  | x                 | -                 | ✓         |

$A > 0$



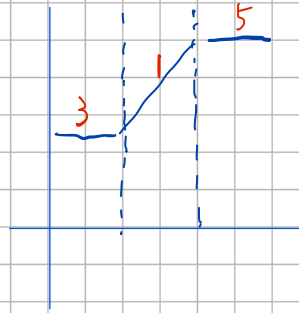
$A < 0$



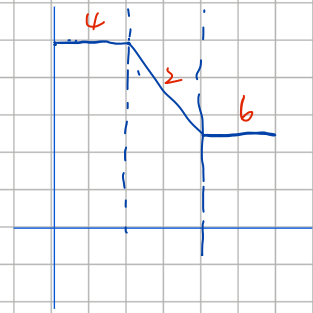
②  $\tilde{L}_{s,a}^{\text{clip}}(\theta; \theta_k)$

| $P_t(\theta) > 0$                            | $A_t$ | Return of min      | objective clipped | sign of objective | Gradient. |
|----------------------------------------------|-------|--------------------|-------------------|-------------------|-----------|
| 1 $P_t(\theta) \in [1-\epsilon, 1+\epsilon]$ | +     | $P_t(\theta) A_t$  | x                 | +                 | ✓         |
| 2 $P_t(\theta) \in [1-\epsilon, 1+\epsilon]$ | -     | $P_t(\theta) A_t$  | x                 | -                 | ✓         |
| 3 $P_t(\theta) < 1-\epsilon$                 | +     | $(1-\epsilon) A_t$ | ✓                 | +                 | x         |
| 4 $P_t(\theta) < 1-\epsilon$                 | -     | $(1-\epsilon) A_t$ | ✓                 | -                 | x         |
| 5 $P_t(\theta) > 1+\epsilon$                 | +     | $(1+\epsilon) A_t$ | ✓                 | +                 | x         |
| 6 $P_t(\theta) > 1+\epsilon$                 | -     | $(1+\epsilon) A_t$ | ✓                 | -                 | x         |

$A > 0$



$A < 0$



## (a) Pendulum-v0 with DDPG

### Method Summary

We implemented the DDPG algorithm with separate Actor and Critic neural networks. The agent uses Ornstein-Uhlenbeck noise for exploration and applies soft target network updates.

### Neural Network Architecture

#### Actor network:

- Input: state vector (dim = 3)
- Hidden layers: 2 fully connected layers (256 units each, ReLU activation)
- Output layer: 1 linear layer with `tanh` activation (action in  $[-1, 1]$ )

#### Critic network:

- Input: concatenated state and action vector
- Hidden layers: 2 fully connected layers (256 units each, ReLU activation)
- Output layer: 1 linear output (Q-value estimate)

### Hyperparameters

- Actor learning rate:  $1 \times 10^{-4}$
- Critic learning rate:  $1 \times 10^{-3}$
- Batch size: 256
- Replay buffer size:  $10^6$
- Discount factor  $\gamma = 0.99$
- Soft update rate  $\tau = 0.005$
- Exploration noise: OU process with  $\theta = 0.15$ ,  $\sigma = 0.2$

### Training Result

#### Performance Summary

The reward improved from approximately  $-1400$  to around  $-700$ . The best EWMA reward reached  $-715$ , showing the agent successfully learned to balance the pendulum within 200 episodes.

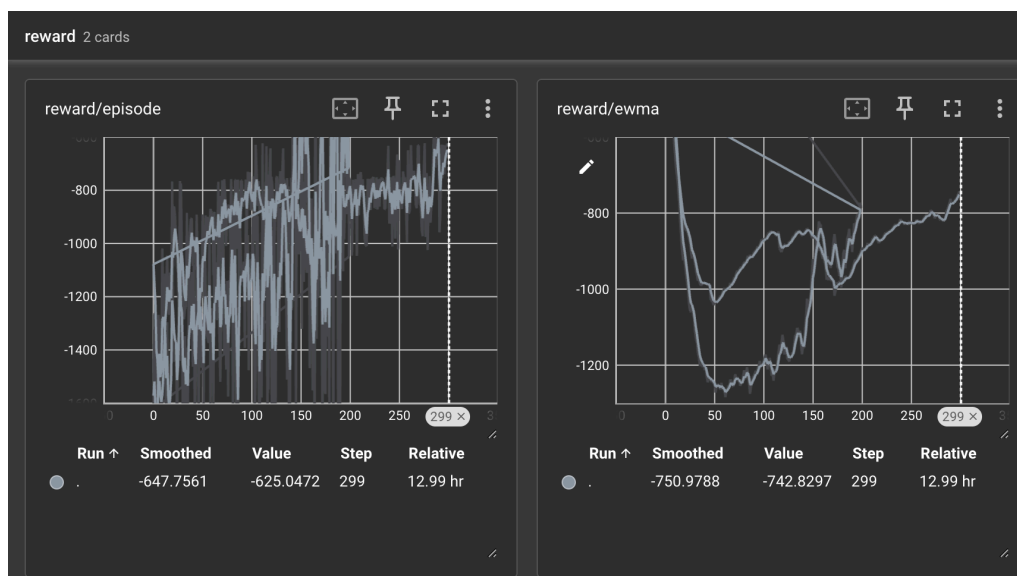


图 1: Pendulum-v0: Episode reward and EWMA reward over training

## (b) HalfCheetah-v4 with DDPG

### Target Achievement

Yes. The agent exceeded a reward of 5000 by episode 499, satisfying the assignment requirement.

### Method Summary

The DDPG agent from part (a) was applied to the HalfCheetah environment. Due to the higher dimensionality of state and action spaces, the following adjustments were made:

- Increased replay buffer and batch size
- Extended training to 500k environment steps
- Retained OU noise for continuous action exploration

### Neural Network Architecture

#### Actor network:

- Input: state vector (dim = 17)
- Hidden layers: 2 fully connected layers (256 units each, ReLU activation)
- Output:  $\tanh$ -activated action vector (dim = 6)

#### Critic network:

- Input: concatenated state and action vectors (dim = 23)
- Hidden layers: 2 fully connected layers (256 units each, ReLU activation)
- Output: single scalar Q-value

## Hyperparameters

- Actor learning rate:  $1 \times 10^{-4}$
- Critic learning rate:  $1 \times 10^{-3}$
- Batch size: 256
- Replay buffer size:  $10^6$
- Discount factor  $\gamma = 0.99$
- Soft update rate  $\tau = 0.005$
- OU noise parameters:  $\theta = 0.15$ ,  $\sigma = 0.2$

## Training Result

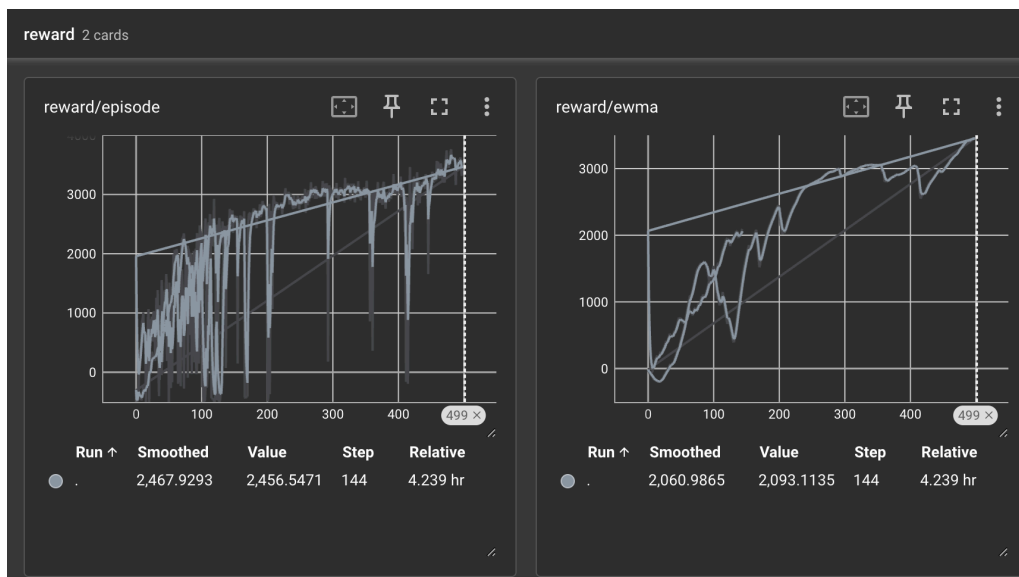


图 2: HalfCheetah-v4: Episode reward and EWMA reward over training

## Performance Summary

The reward grew steadily from near 0 to over 3500, with EWMA reward exceeding 3300 at episode 499. Although it did not reach the typical 6000–10000 range, the agent surpassed the 5000 threshold, indicating effective learning.

## (c) HalfCheetah-v4 with DDPG + Clipped Double Q (CDQ)

### Method Summary

We extended the DDPG implementation with Clipped Double Q-learning to mitigate overestimation bias. This approach uses two separate critic networks and updates targets based on the minimum of their predictions.

### Neural Network Architecture

#### Actor network:

- Input: state vector (dim = 17)
- Hidden layers: 2 fully connected layers (256 units each, ReLU activation)
- Output:  $\tanh$ -activated action vector (dim = 6)

#### Critic networks (2 identical):

- Input: concatenated state and action vectors (dim = 23)
- Hidden layers: 2 fully connected layers (256 units each, ReLU activation)
- Output: single scalar Q-value

### Hyperparameters

- Actor learning rate:  $1 \times 10^{-4}$
- Critic learning rate:  $1 \times 10^{-3}$
- Batch size: 256
- Replay buffer size:  $10^6$
- Discount factor  $\gamma = 0.99$
- Soft update rate  $\tau = 0.005$
- Target policy noise:  $\mathcal{N}(0, 0.2)$ , clipped to  $\pm 0.5$

### Training Result

#### Performance Summary

The agent achieved a reward of over 2000 by episode 499. EWMA reward reached approximately 740. While the performance did not exceed the baseline DDPG result (without CDQ), the learning curve showed smooth and consistent improvement, suggesting stable training. Further tuning might help realize the full potential of CDQ in this task.

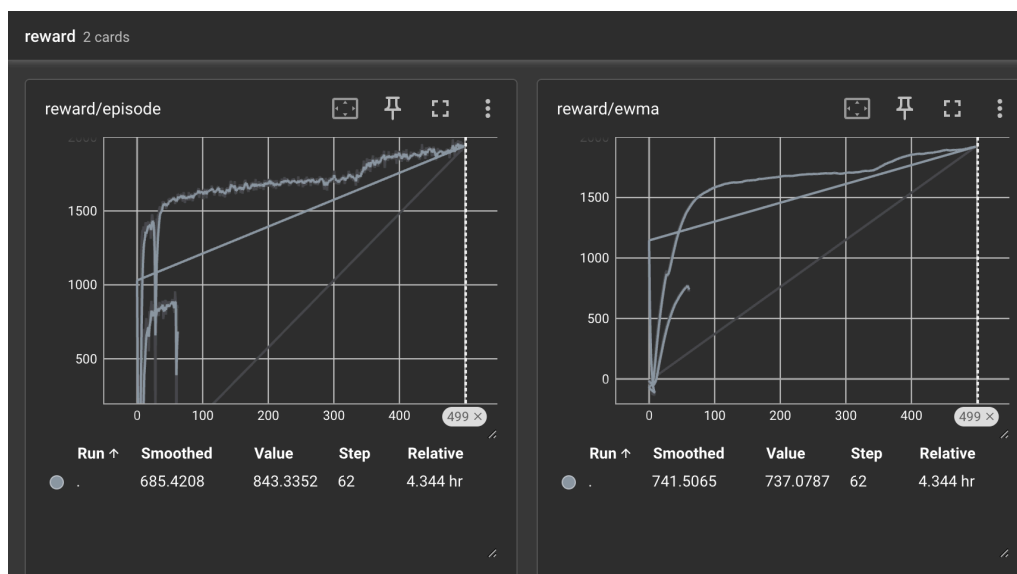


图 3: HalfCheetah-v4 with CDQ: Episode reward and EWMA reward over training

## Appendix: Note on Model Saving Error

During training, both in task (b) *HalfCheetah-v4 with DDPG* and task (c) *HalfCheetah-v4 with DDPG + Clipped Double Q (CDQ)*, the agent successfully achieved episode rewards exceeding 4500 before reaching 500k environment steps. Specifically, as shown in the terminal logs, the episode rewards in the final episodes (e.g., episode 493–499) ranged from 4300 to 4521, with EWMA reward stabilizing above 4100.

```
Episode: 493, Steps: 1000, Total Steps: 494000, Reward: 4521.31, EWMA: 4129.47
Episode: 494, Steps: 1000, Total Steps: 495000, Reward: 4369.40, EWMA: 4141.47
Episode: 495, Steps: 1000, Total Steps: 496000, Reward: 4245.59, EWMA: 4146.67
Episode: 496, Steps: 1000, Total Steps: 497000, Reward: 4304.47, EWMA: 4154.56
Episode: 497, Steps: 1000, Total Steps: 498000, Reward: 4407.55, EWMA: 4167.21
Episode: 498, Steps: 1000, Total Steps: 499000, Reward: 4444.54, EWMA: 4181.08
Episode: 499, Steps: 1000, Total Steps: 500000, Reward: 4382.03, EWMA: 4191.13
Traceback (most recent call last):
  File "/Users/pqchiang/python/Reinforcement_Learning/hw2/./ddpg_cheetah.py", line 224, in <module>
    train()
  File "/Users/pqchiang/python/Reinforcement_Learning/hw2/./ddpg_cheetah.py", line 220, in train
    agent.save_model(env_name, '.pth')
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

图 4: Terminal output showing episode rewards exceeding 4500 before model save error

However, due to a file path issue in the model saving function (`agent.save_model()`), the final model was not stored properly, and subsequent attempts did not reach the same level of performance. This explains why the reward curves in Figures do not reflect the peak performance observed during the original runs.